

Week 11 - Lab Materials – Instructions:

Beech Hill Library Project - Project Aims:

Beech Hill (primary and secondary) school has 800 students and a teaching staff of 50.

The library is open to the students all day during school hours and is currently operated by the English Department teaching staff on a rotation basis.

Providing an engaging library experience for students with diverse age, skill and competency levels is not easy. In installing a new library system it is envisaged that the students will be able to enjoy an intuitive, friendly and effective interaction with the wealth of information the library has to offer. This software solution must enable the library provide students with a positive and educational library experience.

The new library system will streamline library operations and improve the day-to-day efficiency of the staff. The school also wants to provide students, parents and educators with an affordable, easy-to-use, integrated software solution which will provide for effective management of the library. The solution will support the teachers and learners alike in making the most of the available learning resources - with access to information leading to learning and achievement.

The school require a computer based system that all teachers will be able to use and utilise in helping their students' access all library items. Records of books, multimedia materials and the like are tracked through the system to determine usage, availability and requests to reserve items.

The main functions of the system are outlined in the diagram below:

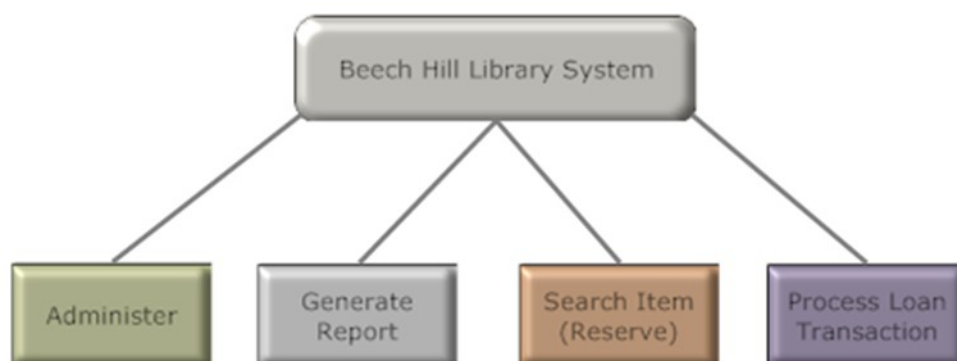


Figure 1- Beech Hill Library System Main Functions

Scenario:

Work in pairs - Work through the following tasks together.

Note: If the task doesn't specify the role that should work on the task then you should both work on it and should do so from the perspective of the test analyst.

Task 1.

You are both working as developers and have been working on the code for checking the age of students and want to do some data flow analysis on the code that has been developed. Explain what data flow analysis is useful for and explain the three types of anomalies that this type of analysis may identify.

Task 2.

You are both working as developers and have written some basic code to check if a student is under or over 15 years of age using an If-then- else structure. Prepare a control flow graph for this piece of code and explain clearly how you would use this diagram in the context of path testing. Your manager walks over while you are drawing the diagram and asks what you are doing. Explain to them how this approach may be useful using a more complex code example of your choosing and explain clearly the link between the control flow diagram and the approach to testing based on path selection.

Task 3.

You are both working as developers and your manager wants you to use "Test Driven Development" for the Beech Hill Library Project. They have asked that you prepare a report for the Beech Hill Library Project Stakeholders that clearly explains what this approach entails. The report should also contain an explanation of the benefits of using this approach.

Task 5.

You are both working as developers and have been tasked with planning the writing of unit tests for the system. Explain the approach that you would take in writing your unit tests and explain where you will locate these test cases and why. What types of test case would you suggest? What tools do you think would be suitable and would they be suitable for any other stages of testing? Should you write test cases for all code that will be written for the system and if so, why/why not?

Task 6.

Using the assert statements that have been discussed in this module, explain each of the assert statements discussed. Explain, using a clear example for each one, what type of test case you would use for each assert statement for.

Task 7.

Explain the difference between debugging and testing. Why, in your opinion, is it important to make this distinction? Using an example from Beech Hill of your choosing or an example discussed in the module, explain clearly what the 5 IDLE debugger options are used for and how they can be useful in locating and then fixing defects.